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= 6pt, backgroundcolor = lightgreen!30, roundcorner = 3pt,

kibr  $k_{ibr}$  IBR current ratio,  $I_{IBR}/I_{rated}$  Idiff  $I_{diff}$  Differential operating current Ibias  $I_{bias}$   
Bias (restraint) current Iop  $I_{op}$  Relay operating (pickup) current km  $k_m$  Adaptive bias  
slope of 87L characteristic Zline  $Z_{line}$  Line positive-sequence impedance Zapp  $Z_{app}$   
Apparent impedance seen by distance relay alpha  $\alpha$  Per-unit fault location along  
protected line (0 = local end) TMS **TMS** Time-Multiplier Setting for IEC inverse-time  
overcurrent relay CTI **CTI** Coordination Time Interval between primary and backup  
relay M **M** Relay current multiple,  $I_{fault}/I_{pickup}$  lambda  $\lambda$  Forgetting factor in online  
SGD ( $0 < \lambda \leq 1$ ) S  $S_k$  CUSUM score at sample  $k$  delta  $\delta$  CUSUM reference value  
(allowable slack) theta  $\theta$  CUSUM alarm threshold P **P** EKF error covariance matrix K  
**K** EKF Kalman gain matrix Q **Q** EKF process noise covariance R **R** EKF  
measurement noise covariance



# **CHAPTER 1**

## **TEST CHAPTER**

Hello world. This is a test.

